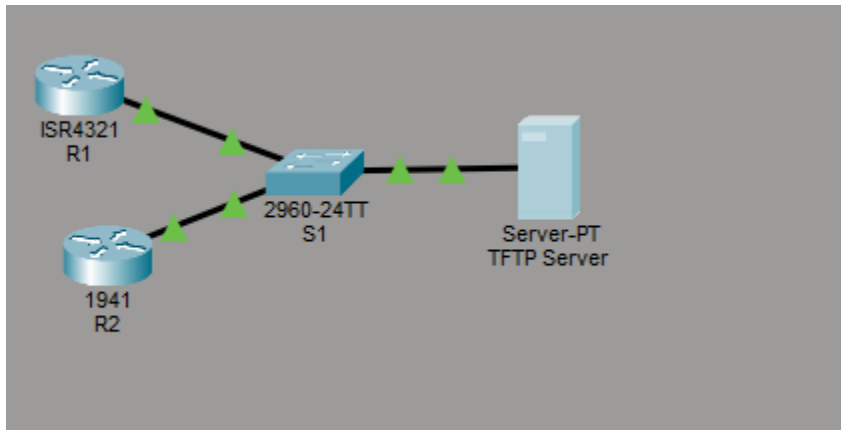


lab16 10.7.6 Packet Tracer - Use a TFTP Server to Upgrade a Cisco IOS Image - ILM

นายจิรภัทร แก้วดี 673380577-9 Sec 2

topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0	192.168.2.1	255.255.255.0	N/A
R2	G0/0	192.168.2.2	255.255.255.0	N/A
S1	VLAN 1	192.168.2.3	255.255.255.0	192.168.2.1
TFTP Server	NIC	192.168.2.254	255.255.255.0	192.168.2.1

Objectives

Part 1: Upgrade an IOS Image on a Cisco Device

Part 2: Backup an IOS Image on a TFTP Server

Scenario

A TFTP server can help manage the storage of IOS images and revisions to IOS images. For any network, it is good practice to keep a backup copy of the Cisco IOS Software image in case the system image in the router becomes corrupted or accidentally erased. A TFTP server can also be used to store new upgrades to the IOS and then deployed throughout the network where it is needed. In this activity, you will upgrade the IOS images on Cisco devices by using a TFTP server. You will also backup an IOS image with the use of a TFTP server.

Instructions

Part 1: Upgrade an IOS Image on a Cisco Device

Step 1: Upgrade an IOS image on a router.

- a. Access the TFTP server and enable the TFTP service.
- b. Note the IOS image files that are available on the TFTP server.

Which IOS images stored on the server are compatible with a 1941 router?

```
asa842-k8.bin
asa923-k8.bin
c1841-advipservicesk9-mz.124-15.T1.bin
c1841-ipbase-mz.123-14.T7.bin
c1841-ipbasek9-mz.124-12.bin
c1900-universalk9-mz.SPA.151-4.M4.bin
c1900-universalk9-mz.SPA.155-3.M4a.bin
c2600-advipservicesk9-mz.124-15.T1.bin
c2600-i-mz.122-28.bin
c2600-ipbasek9-mz.124-8.bin
c2800nm-advipservicesk9-mz.124-15.T1.bin
c2800nm-advipservicesk9-mz.151-4.M4.bin
c2800nm-ipbase-mz.123-14.T7.bin
c2800nm-ipbasek9-mz.124-8.bin
c2900-universalk9-mz.SPA.155-3.M4a.bin
c2950-i6q4l2-mz.121-22.EA4.bin
c2950-i6q4l2-mz.121-22.EA8.bin
c2960-lanbase-mz.122-25.FX.bin
c2960-lanbase-mz.122-25.SEE1.bin
c2960-lanbasek9-mz.150-2.SE4.bin
c3560-advipservicesk9-mz.122-37.SE1.bin
c3560-advipservicesk9-mz.122-46.SE.bin
c800-universalk9-mz.SPA.152-4.M4.bin
c800-universalk9-mz.SPA.154-3.M6a.bin
cat3k_caa-universalk9.16.03.02.SPA.bin
cgr1000-universalk9-mz.SPA.154-2.CG
cgr1000-universalk9-mz.SPA.156-3.CG
ir800-universalk9-bundle.SPA.156-3.M.bin
ir800-universalk9-mz.SPA.155-3.M
ir800-universalk9-mz.SPA.156-3.M
ir800_yocto-1.7.2.tar
ir800_yocto-1.7.2_python-2.7.3.tar
pt1000-i-mz.122-28.bin
pt3000-i6q4l2-mz.121-22.EA4.bin
```


-
- e. Verify that the IOS image has been copied to flash.

```
R2#show flash:

System flash directory:
File Length Name/status
  3  33591768 c1900-universalk9-mz.SPA.151-4.M4.bin
  5  33591768 c1900-universalk9-mz.SPA.155-3.M4a.bin
  2   28282  sigdef-category.xml
  1  227537  sigdef-default.xml
[67439355 bytes used, 188304645 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

- f. R2#

How many IOS images are located in flash?

2

- g. Use the **boot system** command to load the version 15.5 IPBase image on the next reload.

```
R2(config)# boot system flash c1900-universalk9-mz.SPA.155-3.M4a.bin
```

- h. Save the configuration and reload **R2**.

```

R2#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by Cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test

Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
##### [OK]
Smart Init is enabled
smart init is sizing iomem

      TYPE          MEMORY_REQ
Onboard devices &
  buffer pools      0x01E8F000
-----
TOTAL:              0x01E8F000
Rounded IOMEM up to: 32Mb.
Using 6 percent iomem. [32Mb/512Mb]

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(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.
    cisco Systems, Inc.
    170 West Tasman Drive
    San Jose, California 95134-1706

Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.5(3)M4a, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2016 by Cisco Systems, Inc.
Compiled Thu 06-Oct-16 13:56 by mnguyen

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```

แสดง log การ boot ของ R2 หลังสั่ง reload กำลังโหลด IOS image ใหม่ (Version 15.5(3)M4a)

- i. Use the **show version** command to verify the upgraded IOS image is loaded after **R2** reboots.

R2# **show version**

```
R2>show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.5(3)M4a, RELEASE SOFTWARE (f
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2016 by Cisco Systems, Inc.
Compiled Thu 06-Oct-16 13:56 by mnguyen
```

```
ROM: System Bootstrap, Version 15.0(1r)M9, RELEASE SOFTWARE (fc1)
```

```
R2 uptime is 1 minutes, 59 seconds
System returned to ROM by power-on
System image file is "flash0:c1900-universalk9-mz.SPA.155-3.M4a.bin"
Last reload type: Normal Reload
Last reload reason: Reload Command
```

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```
Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
1 terminal line
1 Virtual Private Network(VPN) Module
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
```

License Info:

License UDI:

Device#	PID	SN
*0	CISC01941/K9	FTX1524EK5X-

Suite License Information for Module:'c1900'

Suite	Suite Current	Type	Suite Next reboot
FoundationSuiteK9	None	None	None
securityk9			
datak9			

Suite License Information for Module:'c1900'

Suite	Suite Current	Type	Suite Next reboot
FoundationSuiteK9	None	None	None
securityk9			
datak9			

Technology Package License Information for Module:'c1900'

Technology	Technology-package Current	Technology-package Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	disable	None	None
data	disable	None	None

Configuration register is 0x2102

คำสั่ง show version ขึ้นย่นว่า R2 boot ด้วย IOS 15.5(3)M4a และไฟล์ system image คือ flash0:c1900-universalk9-mz.SPA.155-3.M4a.bin ดำเนินการตามต้องการ

Step 2: Upgrade an IOS image on a switch.

- a. Access the TFTP server and copy the **c2960-lanbasek9-mz.150-2.SE4.bin** image to S1.

S1# **copy tftp: flash:**

```
S1>
S1#enable
S1#copy tftp: flash:
Address or name of remote host []? 192.168.2.254
Source filename []? c2960-lanbasek9-mz.150-2.SE4.bin
Destination filename [c2960-lanbasek9-mz.150-2.SE4.bin]?

Accessing tftp://192.168.2.254/c2960-lanbasek9-mz.150-2.SE4.bin....
Loading c2960-lanbasek9-mz.150-2.SE4.bin from 192.168.2.254: !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 4670455 bytes]

4670455 bytes copied in 3.083 secs (121792 bytes/sec)
```

คำสั่ง copy tftp: flash: บน S1 คัดลอกไฟล์ c2960-lanbasek9-mz.150-2.SE4.bin จาก TFTP server มาสำเร็จ
ขนาด 4,670,455 bytes

- b. Use the **boot system** command to configure the switch to load the new IOS image on boot.
c. Reload S1 and verify the new image has been loaded into memory.

```

S1#reload
Proceed with reload? [confirm]
C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(25r)FX, RELEASE SOFTWARE (fc4)
Cisco WS-C2960-24TT (RC32300) processor (revision C0) with 21039K bytes of memory.
2960-24TT starting...
Base ethernet MAC Address: 0060.2FD5.632E
Xmodem file system is available.
Initializing Flash...
flashfs[0]: 3 files, 0 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 64016384
flashfs[0]: Bytes used: 9342038
flashfs[0]: Bytes available: 54674346
flashfs[0]: flashfs fsck took 1 seconds.
...done Initializing Flash.

Boot Sector Filesystem (bs:) installed, fsid: 3
Parameter Block Filesystem (pb:) installed, fsid: 4

Loading "flash:/c2960-lanbasek9-mz.150-2.SE4.bin"...
##### [OK]
C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(25r)FX, RELEASE SOFTWARE (fc4)
Cisco WS-C2960-24TT (RC32300) processor (revision C0) with 21039K bytes of memory.
2960-24TT starting...
Base ethernet MAC Address: 0060.2FD5.632E
Xmodem file system is available.
Initializing Flash...
flashfs[0]: 3 files, 0 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 64016384
flashfs[0]: Bytes used: 9342038
flashfs[0]: Bytes available: 54674346
flashfs[0]: flashfs fsck took 1 seconds.
...done Initializing Flash.

Boot Sector Filesystem (bs:) installed, fsid: 3
Parameter Block Filesystem (pb:) installed, fsid: 4

Loading "flash:/c2960-lanbasek9-mz.150-2.SE4.bin"...
##### [OK]
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Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.
    cisco Systems, Inc.
    170 West Tasman Drive
    San Jose, California 95134-1706
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
Initializing flashfs...
fsck: Disable shadow buffering due to heap fragmentation.
flashfs[2]: 2 files, 1 directories
flashfs[2]: 0 orphaned files, 0 orphaned directories
flashfs[2]: Total bytes: 32514048

```

Log แสดงการ boot ของ S1 หลังสั่ง reload โดยสามารถโหลดไฟล์ flash:/c2960-lanbasek9-mz.150-2.SE4.bin (IOS ตัวใหม่ Version 15.0(2)SE4) ได้สำเร็จ ไม่มี error

```

S1>show version
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen

ROM: Bootstrap program is C2960 boot loader
BOOTLDR: C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(25r)FX, RELEASE SOFTWARE (fc4)

Switch uptime is 39 minutes
System returned to ROM by power-on
System image file is "flash:c2960-lanbasek9-mz.150-2.SE4.bin"

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cisco WS-C2960-24TT-L (PowerPC405) processor (revision B0) with 65536K bytes of memory.
Processor board ID FOC1010X104
Last reset from power-on
1 Virtual Ethernet interface
24 FastEthernet interfaces
2 Gigabit Ethernet interfaces
The password-recovery mechanism is enabled.

64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address      : 00:60:2F:D5:63:2E
Motherboard assembly number    : 73-10390-03
Power supply part number       : 341-0097-02
Motherboard serial number      : FOC10093R12
Power supply serial number     : AZS1007032H
Model revision number          : B0
Motherboard revision number    : B0
Model number                   : WS-C2960-24TT-L
System serial number           : FOC1010X104
Top Assembly Part Number       : 800-27221-02
Top Assembly Revision Number   : A0
Version ID                    : V02
CLEI Code Number               : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version  SW Image
-----
*    1 26    WS-C2960-24TT-L    15.0(2)SE4    C2960-LANBASEK9-M

Configuration register is 0xF

```

คำสั่ง show version บน S1 ยืนยันว่า:

- Version เปลี่ยนเป็น **15.0(2)SE4** (จากไฟล์ c2960-lanbasek9-mz.150-2.SE4.bin)
- System image file คือ flash:c2960-lanbasek9-mz.150-2.SE4.bin ตรงตามที่ต้องการ

- d. Close the TFTP configuration window if it is still open.

Part 2: Backup an IOS Image to a TFTP Server

- a. On **R1**, display the contents of flash and record the IOS image.

isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin

```
R1# show flash:
```

```
R1>
R1>show flash:

System flash directory:
File      Length      Name/status
  4      4670455  c2960-lanbasek9-mz.150-2.SE4.bin
  3      486899872  isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin
  2       28282   sigdef-category.xml
  1       227537   sigdef-default.xml
[491826146 bytes used, 2757223454 available, 3249049600 total]
3.17338e+06K bytes of processor board System flash (Read/Write)
```

R1>

- b. Use the **copy** command to back up the IOS image in flash memory on **R1** to a TFTP server. **Note:** The **isr4300** image is considerably larger than the **c1900** image. It will take longer to transmit it to the TFTP server.

```
R1# copy flash: tftp:
```

```
Source filename []? isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin
```

```
Address or name of remote host []? 192.168.2.254
```

Destination filename [isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin]?

[illegible]

```
486899872 bytes copied in 18.815 secs (83367 bytes/sec)
```

```
asa842-k8.bin
asa923-k8.bin
c1841-advipservicesk9-mz.124-15.T1.bin
c1841-ipbase-mz.123-14.T7.bin
c1841-ipbasek9-mz.124-12.bin
c1900-universalk9-mz.SPA.151-4.M4.bin
c1900-universalk9-mz.SPA.155-3.M4a.bin
c2600-advipservicesk9-mz.124-15.T1.bin
c2600-i-mz.122-28.bin
c2600-ipbasek9-mz.124-8.bin
c2800nm-advipservicesk9-mz.124-15.T1.bin
c2800nm-advipservicesk9-mz.151-4.M4.bin
c2800nm-ipbase-mz.123-14.T7.bin
c2800nm-ipbasek9-mz.124-8.bin
c2900-universalk9-mz.SPA.155-3.M4a.bin
c2950-i6q4l2-mz.121-22.EA4.bin
c2950-i6q4l2-mz.121-22.EA8.bin
c2960-lanbase-mz.122-25.FX.bin
c2960-lanbase-mz.122-25.SEE1.bin
c2960-lanbasek9-mz.150-2.SE4.bin
c3560-advipservicesk9-mz.122-37.SE1.bin
c3560-advipservicesk9-mz.122-46.SE.bin
c800-universalk9-mz.SPA.152-4.M4.bin
c800-universalk9-mz.SPA.154-3.M6a.bin
cat3k_caa-universalk9.16.03.02.SPA.bin
cgr1000-universalk9-mz.SPA.154-2.CG
cgr1000-universalk9-mz.SPA.156-3.CG
ir800-universalk9-bundle.SPA.156-3.M.bin
ir800-universalk9-mz.SPA.155-3.M
ir800-universalk9-mz.SPA.156-3.M
ir800_yocto-1.7.2.tar
ir800_yocto-1.7.2_python-2.7.3.tar
isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin
pt1000-i-mz.122-28.bin
pt3000-i6q4l2-mz.121-22.EA4.bin
```

ยืนยัน IOS versionยืนยัน IOS version

```
R2>show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.5(3)M4a, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2016 by Cisco Systems, Inc.
Compiled Thu 06-Oct-16 13:56 by mnguyen

ROM: System Bootstrap, Version 15.0(1r)M9, RELEASE SOFTWARE (fc1)

R2 uptime is 13 minutes, 37 seconds
System returned to ROM by power-on
System image file is "flash0:c1900-universalk9-mz.SPA.155-3.M4a.bin"
Last reload type: Normal Reload
Last reload reason: Reload Command

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Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
1 terminal line
1 Virtual Private Network(VPN) Module
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:
License UDI:

-----
Device# PID SN
-----
*0 CISC01941/K9 FTX1524EK5X-

R2>
R2>
R2>enable
R2#show running-config | include boot system
boot system flash c1900-universalk9-mz.SPA.155-3.M4a.bin
R2#
```

คำสั่ง show running-config | include boot system บน R2 ยืนยันว่ามีบรรทัด boot system flash c1900-universalk9-mz.SPA.155-3.M4a.bin บันทึกอยู่จริง

เชื่อว่า flash ยังมีไฟล์ครบ

```
System flash directory:
File Length Name/status
 3 33591768 c1900-universalk9-mz.SPA.151-4.M4.bin
 5 33591768 c1900-universalk9-mz.SPA.155-3.M4a.bin
 2 28282 sigdef-category.xml
 1 227537 sigdef-default.xml
[67439355 bytes used, 188304645 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

R2#

แสดงว่ามี IOS image 2 ไฟล์อยู่ใน flash ของ R2 (ไฟล์เก่าและไฟล์ใหม่) ครบถ้วน

เช็ค connectivity หลัง reload ว่ายังปกติ

```
R2#ping 192.168.2.254
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.254, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R2#ping 192.168.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.2.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/1/3 ms

R2#
```

Ping ไปยัง TFTP server (192.168.2.254) และ default gateway (192.168.2.1) สำเร็จ 80% (4/5) ยืนยันว่า network connectivity ยังใช้งานได้ปกติหลัง reload

เช็คสถานะ interface

```
R2#
R2#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       192.168.2.2    YES NVRAM  up                    up
GigabitEthernet0/1       unassigned      YES NVRAM  administratively down down
Vlan1                    unassigned      YES NVRAM  administratively down down
R2#
```

show ip interface brief ยืนยันว่า G0/0 ยังมี IP 192.168.2.2 และสถานะ up/up ตามปกติ